Predict a position and a range

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict a future position and explain the limits of the prediction.

SUCCESS CRITERIA - I can...

1. Use matching units and include the starting position.
2. Report a central prediction and its bounds.
3. Explain how an uncertain measurement differs from a limit of the calculation.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “initial” and “final” to explain your answer. Sentence starters are optional.

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

bound: an upper or lower limit in a model

01 Warm-up: bring back an earlier idea

Start at **5 m**, move to **14 m**, then finish at **7 m**.
1. **Draw the route**.
2. **Find distance**, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $x_f = x_i + v\,\Delta t$.
Constant velocity; i = initial, f = final. Use the same units for all positions.

Worked reasoning: A cart starts at **1 m** and moves right at **2 m/s** for **3 s**. Central prediction: **1**+**2**×**3**=**7 m**. If its speed could be **1** to **3 m/s**, the endpoint range is **4** to **10 m**.

Your turn: A cart moves along a marked track.
1. **Show its start**, central predicted endpoint, and full possible endpoint range on the number line.
2. What must remain constant throughout the **3 s**?

Sentence starter: I predict ... . I used ... to make this prediction because ... .



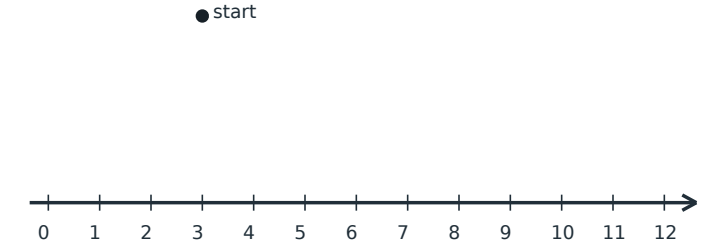
Position (m); right is positive

Marzano check - circle: ① ② ③

03 Start plus change

A cart starts at **3 m** and moves right at **2 m/s** for **4 s**. **Find its displacement** and final position. **Mark the endpoint**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Predict leftward motion

A robot starts at the **12 m** mark and drives left at a steady **2 m/s** for **3 s**.
1. **Draw the move** on the number line.
2. **Find the displacement** and final position.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Match the time units

A rover moves at **2 m/s** for **1 minute**.
1. Use **1 minute** = **60 seconds**.
2. **Find its distance** at this steady speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

06 Read the range

A prediction is **10 m** with a range from **8 m** to **12 m**.
1. **Label the middle**, lower end, and upper end.
2. **Find how far each end** is from the middle.

Sentence starter: I predict ... . I used ... to make this prediction because ... .



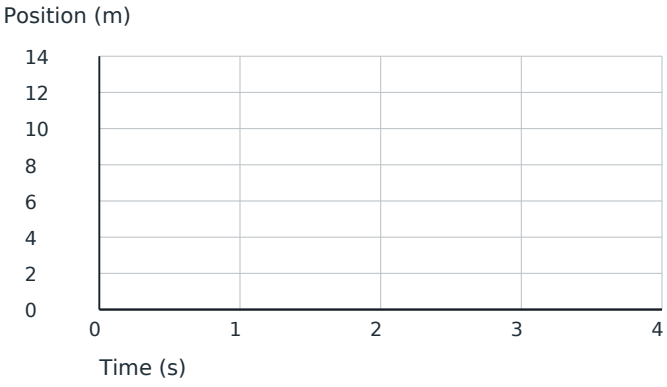
Marzano check - circle: ① ② ③

07 A slow and a fast case

A rover starts at **0 m** and travels right at a constant speed somewhere between **1** and **3 m/s**.
1. **Draw the slowest** and fastest possible position-time lines.
2. **Find the possible position range** after **4 s**.

Support: Slow case: ____. Fast case: ____. My range is ____.

Sentence starter: I found ... [unit]. This value means ... in this situation.



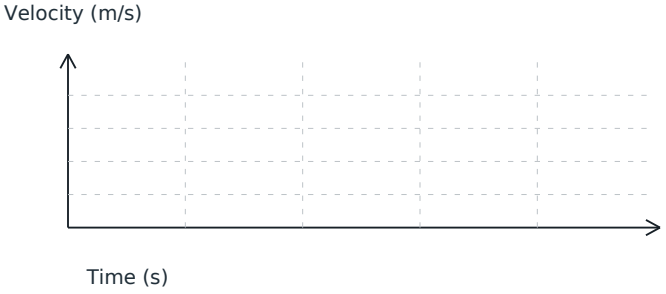
Marzano check - circle: ① ② ③

08 Review: connect an earlier idea

A velocity graph is constant at **-3 m/s** for **5 s**. **Find displacement**. A second object has components **7 m/s** *east* and **2 m/s** *south*; **sketch its direction**. **Sketch the velocity**–time graph; **label the given values**.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

09 Exit check

A cart starts at **2 m** and moves right at **3 m/s** for **2 s**. **Find the central final position**. If the possible final positions run from **7** to **9 m**, state the range and one thing you are treating as true.

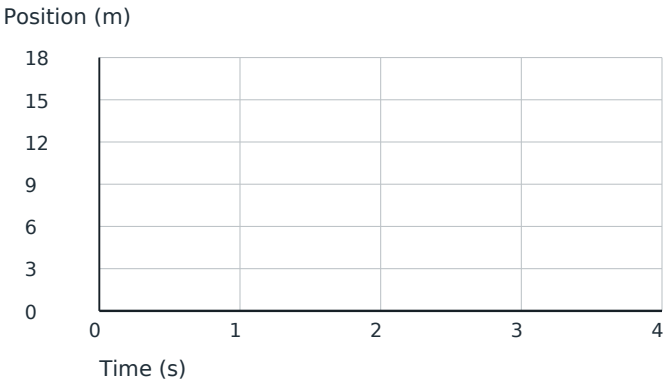
Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

Marzano check - circle: ① ② ③

10 STRETCH: test the prediction

1. Design two different starting-position and speed pairs that both give an endpoint of **10 m** after **2 s**.
2. **Draw the routes**.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③